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Program: BTech CSE (B16)

HDOC Day-12

Menu-Driven Program Using Switch-Case

Problem Statement

Write a menu-driven C program using `switch-case` to input a **positive integer** and perform one of the following operations:

1. Count the number of **even digits and odd digits**.
 2. Find the **sum of even digits and sum of odd digits separately**.
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Algorithm

Step 1: Start

Step 2: Input

Read a positive integer `num`.

Step 3: Display Menu

Display:

1. Count even and odd digits
2. Find sum of even and odd digits

Step 4: Input Choice

Read the user's choice in `choice`.

Step 5: Use Switch-Case

Case 1: Count even and odd digits

1. Initialize:
 - o `even = 0`
 - o `odd = 0`
2. Repeat while `num > 0`:
 - o Find last digit: `digit = num % 10`
 - o If `digit % 2 == 0`, increment `even`.
 - o Otherwise, increment `odd`.
 - o Remove last digit: `num = num / 10`
3. Display `even` and `odd`.

Case 2: Sum even and odd digits

1. Initialize:
 - o `sum_even = 0`
 - o `sum_odd = 0`
2. Repeat while `num > 0`:
 - o Find last digit: `digit = num % 10`
 - o If `digit % 2 == 0`, add it to `sum_even`.
 - o Otherwise, add it to `sum_odd`.
 - o Remove last digit: `num = num / 10`
3. Display `sum_even` and `sum_odd`.

Default Case:

Display "Invalid choice".

Step 6: Stop

Test Case Table

Test Case	Number Choice		Expected Output	Actual Output	Result
1	12045	1	Even_dig = 3, Odd_dig = 2	Even_dig = 3, Odd_dig = 2	Pass
2	12345	2	Sum_even = 6, Sum_odd = 9	Sum_even = 6, Sum_odd = 9	Pass
3	24681	1	Even_dig = 4, Odd_dig = 1	Even_dig = 4, Odd_dig = 1	Pass
4	24681	2	Sum_even = 20, Sum_odd = 1	Sum_even = 20, Sum_odd = 1	Pass

Note: 0 is an **even digit**, so in 12045, the even digits are 2, 0, 4 → **3 even digits**.

C Program

```
#include <stdio.h>

int main()
{
    int num, choice, digit;
    int even = 0, odd = 0;
    int sum_even = 0, sum_odd = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    printf("\n1. Count even and odd digits");
    printf("\n2. Find sum of even and odd digits");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            while(num > 0)
            {
                digit = num % 10;

                if(digit % 2 == 0)
                    even++;
                else
                    odd++;

                num = num / 10;
            }

            printf("Even_dig = %d\n", even);
            printf("Odd_dig = %d\n", odd);
            break;

        case 2:
            while(num > 0)
            {
                digit = num % 10;

                if(digit % 2 == 0)
                    sum_even = sum_even + digit;
                else
                    sum_odd = sum_odd + digit;

                num = num / 10;
            }
    }
}
```

```

        printf("Sum_even = %d\n", sum_even);
        printf("Sum_odd = %d\n", sum_odd);
        break;

    default:
        printf("Invalid choice\n");
}

return 0;
}

```

Compilation and Execution

Using the `cc` compiler in Kali Linux:

```
cc day12.c
./a.out
```

```

(kali㉿kali)-[~/Desktop]
$ cc day12.c

(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter a positive integer: 12045

1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Even_dig = 3
Odd_dig = 2

```

```

(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter a positive integer: 12345

1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 2
Sum_even = 6
Sum_odd = 9

```

```

(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter a positive integer: 24681

1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Even_dig = 4
Odd_dig = 1

```

```

(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter a positive integer: 24681

1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 2
Sum_even = 20
Sum_odd = 1

```

Example — Choice 1

```

Enter a positive integer: 12045

1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1

Even_dig = 3
Odd_dig = 2

```

Example — Choice 2

Enter a positive integer: 12345

1. Count even and odd digits
2. Find sum of even and odd digits

Enter your choice: 2

Sum_even = 6
Sum_odd = 9

Evaluation Against Prepared Test Cases

Test Case 1:

Input: 12045, Choice: 1

Digits: 1, 2, 0, 4, 5

Even digits: 2, 0, 4 → 3

Odd digits: 1, 5 → 2

Expected = Actual → **PASS**

Test Case 2:

Input: 12345, Choice: 2

Even digits: 2 + 4 = 6

Odd digits: 1 + 3 + 5 = 9

Expected = Actual → **PASS**

Test Case 3:

Input: 24681, Choice: 1

Even digits: 2, 4, 6, 8 → 4

Odd digits: 1 → 1

Expected = Actual → **PASS**

Test Case 4:

Input: 24681, Choice: 2

Even sum: 2 + 4 + 6 + 8 = 20

Odd sum: 1

Expected = Actual → **PASS**

Result

The menu-driven C program was successfully implemented using **switch-case** and tested against all prepared test cases. **All test cases passed successfully.**